Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building AI-powered enterprise workflows, responsive interfaces, APIs, and cloud applications. Experienced in React, AngularJS, TypeScript, Python, REST APIs, LLM applications, databases, automated testing, CI/CD, and production delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and responsive interfaces for a multi-tenant AI platform serving 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through indexed queries.
- Implemented AI-assisted workflows for 5,000+ monthly jobs using explicit states, retries, validation, and failure isolation, tripling throughput while improving correctness, observability, maintainability, and reliable production behavior.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed responsive React interfaces and Flask REST APIs handling 10,000+ daily requests, translating instructor requirements into secure upload, routing, status, and conversational grading workflows for 150+ users.
- Created concurrent processing for 1,000+ files with automated unit and integration validation, AWS RDS persistence, and structured error handling, reducing turnaround 60% while maintaining reliable application behavior.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible transformer and data pipelines processing 100,000+ records, applying structured context construction, automated evaluation, instrumentation, and systematic debugging to improve long-horizon planning accuracy 18% across controlled experiments.
- Collaborated on model design, code reviews, and validation experiments, documenting technical decisions and tuning a context-aware architecture that improved closed-loop prediction stability 15% across repeated simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed backend services, REST integrations, and ETL workflows using Python, SQL, AWS Glue, and S3, improving response times 30% and data-processing efficiency 68% across customer-facing enterprise applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across three platforms, accelerating bug fixes, code review feedback, and overall confidence in production deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight secure authentication, permissions, notifications, and user-facing workflows, synchronizing 2 million+ Azure SQL records through maintainable services and real-time WebSockets.
- Automated Docker builds, unit tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, deployment consistency, and operational reliability.

Skills

Languages & Frontend: TypeScript, JavaScript, Python, Java, SQL, React, AngularJS, Next.js, HTML5, CSS3, Tailwind CSS, responsive UI, component design
Backend & Workflows: REST APIs, JSON, Flask, FastAPI, Node.js, microservices, WebSockets, business logic, workflow automation, conversational experiences, API integration
AI & Data: LLM applications, AI-assisted workflows, transformers, model evaluation, FAISS, vector search, NLP, PostgreSQL, Azure SQL, MySQL, MongoDB, AWS RDS, data modeling
Cloud & Quality: AWS, Azure, GCP, Docker, Kubernetes, Git, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, end-to-end testing, code review, debugging, documentation, observability, Agile

Projects

Big-Brain | Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, REST APIs, GCP*
- Built an AI-native document platform using LLaMA-2, FAISS, and REST APIs, delivering conversational search, summarization, and real-time collaborative workflows for 50,000+ documents with 95% query accuracy.

HMIS | Live | *TypeScript, React, REST APIs, Appwrite, Sentry, Twilio*
- Built regulated healthcare workflows for appointments, patient records, and consent-based access using React, REST APIs, authentication, automated notifications, and Sentry monitoring, reducing overall administrative workload 30%.

PiXMod | Live | *AngularJS, Next.js, TypeScript, MongoDB, Stripe, Transformers*
- Developed an AI-assisted full-stack application with AngularJS and Next.js interfaces, REST integrations, transformer workflows, MongoDB persistence, and reusable components, reliably processing 1,000+ user-edited digital media assets.